ISE 555 HW 4

Luke Pepin

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Solution sheet: `HW4.xlsx` (one sheet per question). Pivots are boxed, and the row 0 RHS is $-z$.

Question 1

Minimize: $z = -5x_1 - 7x_2 - 12x_3 + x_4$

Subject to:

- $2x_1 + 3x_2 + 2x_3 + x_4 + s_1 = 38$
- $3x_1 + 2x_2 + 4x_3 - x_4 + s_2 = 55$
- $x_1, x_2, x_3, x_4, s_1, s_2 \geq 0$

Iteration 0

	x_1	x_2	x_3	x_4	s_1	s_2	RHS	ratio
row 0	-5	-7	-12	1	0	0	0	
row 1 (s_1)	2	3	2	1	1	0	38	19
row 2 (s_2)	3	2	$\boxed{4}$	-1	0	1	55	13.75

x_3 enters, s_2 leaves.

Iteration 1

	x_1	x_2	x_3	x_4	s_1	s_2	RHS	ratio
row 0	4	-1	0	-2	0	3	165	
row 1 (s_1)	0.5	2	0	$\boxed{1.5}$	1	-0.5	10.5	7
row 2 (x_3)	0.75	0.5	1	-0.25	0	0.25	13.75	–

x_4 enters, s_1 leaves.

Iteration 2

	x_1	x_2	x_3	x_4	s_1	s_2	RHS
row 0	14/3	5/3	0	0	4/3	7/3	179
row 1 (x_4)	1/3	4/3	0	1	2/3	$-1/3$	7
row 2 (x_3)	5/6	5/6	1	0	1/6	1/6	15.5

Row 0 is all ≥ 0, so this is optimal.

Answer: $x_1 = 0$, $x_2 = 0$, $x_3 = 15.5$, $x_4 = 7$, $z = -179$

Question 2

Minimize: $z' = -5x_1 - 3x_2 - 2x_3$, where $z = -z'$

Subject to:

- $4x_1 + 5x_2 + 2x_3 + x_4 + s_1 = 20$
- $3x_1 + 4x_2 - x_3 + x_4 + s_2 = 30$
- $x_1, x_2, x_3, x_4, s_1, s_2 \geq 0$

Iteration 0

	x_1	x_2	x_3	x_4	s_1	s_2	RHS	ratio
row 0	-5	-3	-2	0	0	0	0	
row 1 (s_1)	4	$\boxed{5}$	2	1	1	0	20	4
row 2 (s_2)	3	4	-1	1	0	1	30	7.5

x_2 enters, s_1 leaves.

Iteration 1

	x_1	x_2	x_3	x_4	s_1	s_2	RHS	ratio
row 0	-2.6	0	-0.8	0.6	0.6	0	12	
row 1 (x_2)	$\boxed{0.8}$	1	0.4	0.2	0.2	0	4	5
row 2 (s_2)	-0.2	0	-2.6	0.2	-0.8	1	14	–

x_1 enters, x_2 leaves.

Iteration 2

	x_1	x_2	x_3	x_4	s_1	s_2	RHS
row 0	0	3.25	0.5	1.25	1.25	0	25
row 1 (x_1)	1	1.25	0.5	0.25	0.25	0	5
row 2 (s_2)	0	0.25	-2.5	0.25	-0.75	1	15

Row 0 is all ≥ 0, so this is optimal.

Answer: $x_1 = 5$, $x_2 = 0$, $x_3 = 0$, $x_4 = 0$, $z = 25$

Question 3

Minimize: $z' = -3x_1 - 9x_2$, where $z = -z'$

Subject to:

- $-5x_1 + 2x_2 + s_1 = 30$
- $-3x_1 + x_2 + s_2 = 12$
- $x_1, x_2, s_1, s_2 \geq 0$

Iteration 0

	x_1	x_2	s_1	s_2	RHS	ratio
row 0	-3	-9	0	0	0	
row 1 (s_1)	-5	2	1	0	30	15
row 2 (s_2)	-3	$\boxed{1}$	0	1	12	12

x_2 enters, s_2 leaves.

Iteration 1

	x_1	x_2	s_1	s_2	RHS	ratio
row 0	-30	0	0	9	108	
row 1 (s_1)	$\boxed{1}$	0	1	-2	6	6
row 2 (x_2)	-3	1	0	1	12	–

x_1 enters, s_1 leaves.

Iteration 2

	x_1	x_2	s_1	s_2	RHS	ratio
row 0	0	0	30	-51	288	
row 1 (x_1)	1	0	1	-2	6	–
row 2 (x_2)	0	1	3	-5	30	–

s_2 should enter, but its column has no positive entries, so there is no ratio test.

Answer: Unbounded, no optimal solution ($z \to \infty$).

Question 4

Minimize: $z' = -7x_1 - 8x_2$, where $z = -z'$

Subject to:

- $4x_1 + x_2 + s_1 = 100$
- $x_1 + x_2 + s_2 = 80$
- $x_1 + s_3 = 40$
- $x_1, x_2, s_1, s_2, s_3 \geq 0$

Iteration 0

	x_1	x_2	s_1	s_2	s_3	RHS	ratio
row 0	-7	-8	0	0	0	0	
row 1 (s_1)	4	1	1	0	0	100	100
row 2 (s_2)	1	$\boxed{1}$	0	1	0	80	80
row 3 (s_3)	1	0	0	0	1	40	–

x_2 enters, s_2 leaves.

Iteration 1

	x_1	x_2	s_1	s_2	s_3	RHS
row 0	1	0	0	8	0	640
row 1 (s_1)	3	0	1	-1	0	20
row 2 (x_2)	1	1	0	1	0	80
row 3 (s_3)	1	0	0	0	1	40

Row 0 is all ≥ 0, so this is optimal.

Answer: $x_1 = 0$, $x_2 = 80$, $z = 640$

Question 5

I am very excited that the company has listened to my suggestion to implement optimization as a way to improve production. As we enter this endeavor, I would like to bring up a few pitfalls that others in situations similar to ours have fallen into. First, the model isn't reality. I'm sure my pitch was fantastic and we want to throw all caution to the wind to solve our issues; however, an optimization is only as good as the mathematical model behind it. We will attempt to improve production to the best of our capabilities, but our solution will be hyper-optimized to the model, and that model is created by us. If we take shortcuts during its creation or take liberties with its variables (or, in some cases, miss them entirely), that will heavily impact our proposed solution, and it may not align with what actually happens on the floor. It is important to begin implementation moderately and cautiously as we learn to model our production environment.

I have also seen Overall Equipment Effectiveness (OEE) cheating in previous work, where production is set up to optimize for the wrong variable. When you optimize for an improper objective, solutions like turning off machines to increase OEE get recommended, so we must be wary of the solutions the model gives us.